

BLK II - UNIT 3: METOC Prediction Models

a. Introduction to METOC Prediction Models

Model Introduction

The goal of this objective will be to show you have the capability to identify the accuracy of the model data using live data as a comparison. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly determine the accuracy and usefulness of the model based on the information discussed throughout this objective. These foundational forecasting concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

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Numerical weather prediction (NWP) models are three-dimensional mathematical projections of the atmosphere for specific location over time. Numerical models can be a valuable tool in the forecasting process. Knowing which weather parameters are available on a model and how to interpret the data will aid a forecaster in developing their forecast. Supercomputers are used to create NWP products using complex algorithms. Hundreds of models are available worldwide and each one has a specific purposes and focus. Within this course we will look at several different types of models, but throughout your career you will gain experience in many more.



Numerical Weather Model Types

Deterministic models

Deterministic models produce a forecast for a specific value, at a specific time for various weather parameters. For example, a deterministic model would forecast temperatures of 23°C at 15Z or pressure at 998 mb at 22Z for a specific location. The creation process begins by pulling in the observed initial state of the atmosphere through various means such as observations, satellite data, and Skew-T soundings to establish a baseline to produce a computer-generated forecast to predict the future state of the atmosphere. Limitations exist due to the inability to gather precise initial conditions because observations are not available at every location. In CONUS there are hundreds available but other countries or continents can have extremely

limited observed data. With a poor initial data, the model is unable to account for all the data gaps and can result in an unreliable finished product.

Point Forecast: Keesler Air Force Base / Biloxi MS
30.4N 88.94W

Last Update: 9:17 am CST Dec 3, 2025

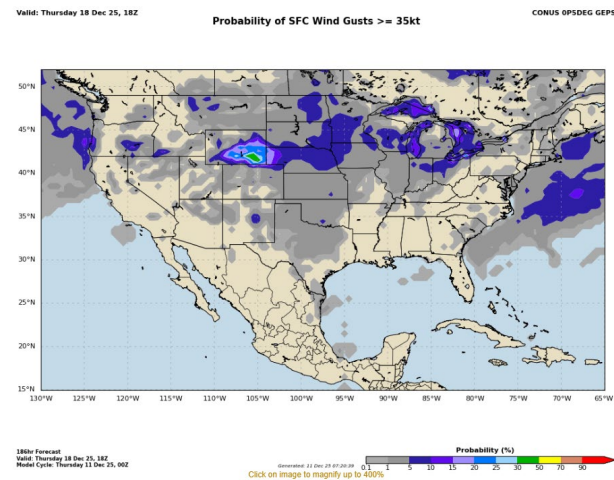
Tabular Forecast

[\[show menu\]](#) [XML](#)

Date	12/03												12/04											
Hour (CST)	11	12	13	14	15	16	17	18	19	20	21	22	23	00	01	02	03	04	05	06	07	08	09	10
Temperature (°F)	54	56	58	59	59	57	55	53	53	51	51	50	50	50	49	48	49	49	49	49	50	52	53	
Dewpoint (°F)	35	36	37	38	38	39	40	40	41	42	42	42	42	42	43	43	43	44	45	46	47	48	48	49
Surface Wind (kt)	5	5	4	4	4	4	3	3	2	2	3	3	3	3	4	4	5	5	6	6	7	9	9	10
Wind Dir	NE	NE	E	E	E	E	E	E	E	E	E	E	E	E	E	NE	NE	NE	NE	NE	NE	NE	NE	NE
Gust																								

Stochastic models

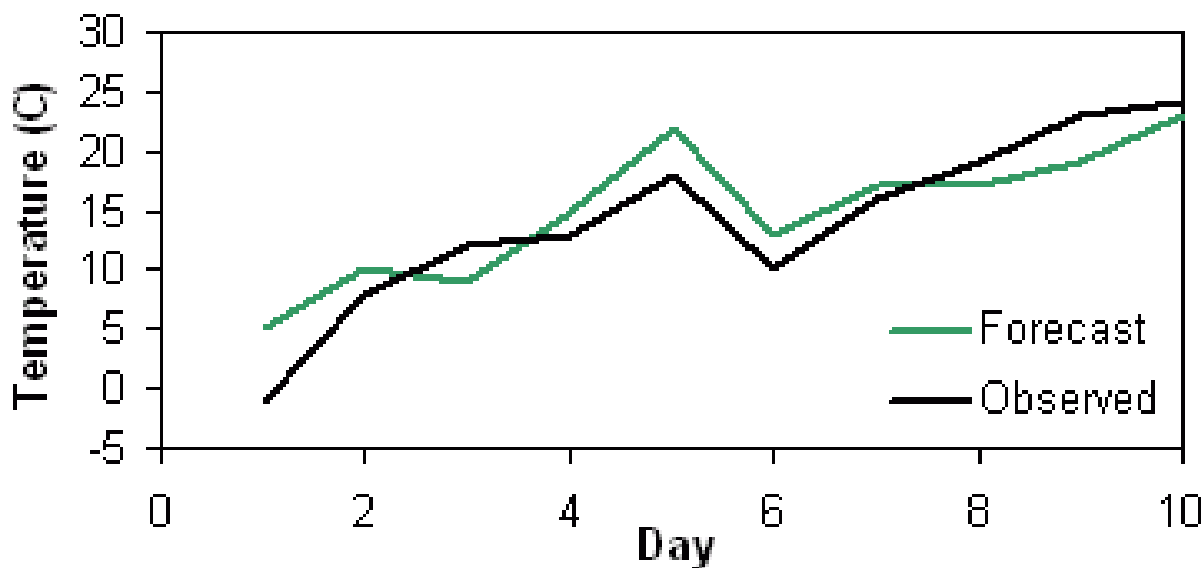
Stochastic models produce a probability of occurrence for various weather parameters. These models would forecast a 20% chance of rain, or a 70% chance of wind speeds greater than or equal to 10KT for a specific location. These models do not offer any insight as to what features are causing the higher probabilities. Stochastic models will need to be supplemented with other model or analysis data to determine the cause of the given probability. The creation process is like deterministic models; however, the algorithm repeats multiple times using different parameters as well as intaking other deterministic model results. The more deterministic models that are forecasting for a specific feature, the higher the probability of occurrence will be on a stochastic model.



Model Verification

Models are an important part of the forecast process; however, they are not infallible. You must continuously evaluate their performance. You must verify the models to figure out their

accuracy. The process requires comparing the model data to the observed data for the specific weather parameters you are forecasting for. Operationally the verification process is becoming more crucial due to the reliance on model data. Model verification has three potential outcomes:



Accurate

The model data matches the observed data, and the information can be used without any adjustments.

Example: You are looking at model data to determine what the days maximum temperature will be. The current observation at 15Z has the temperature at 22°C. The model data says that at 15Z the temperature will be 22°C. This is an accurate temperature forecast, and you can use those temperatures assuming that all the forecast temperatures will also be accurate.

Nearly Accurate

The model data does not match the observed data, but the differences are minor, and the information can be used after making slight adjustments.

Example: You are looking at model data to determine what the pressure will be 6 hours from now. The current observation at 15Z has the pressure at 29.93. The model data says that at 15Z the pressure will be 29.94. This could be considered a nearly accurate pressure forecast, and you can use the pressure values after making slight adjustments. If the model vs. observed pressure was 00.01 too high, you can assume all forecast pressure values will also be off by the same amount. Looking 6 hours ahead, the model is forecasting pressure to be 30.01, you could confidently forecast the actual pressure to be 30.00 after adjusting the values.

Inaccurate

The model data does not mirror the observed data, and the differences make the model unreliable. If a model is deemed inaccurate, you would find a different model that is more accurate.

Example: You are looking at model data to determine what the wind speed and direction will be in 3 hours. The current observation is reporting 35006KT. The model data is forecasting 22012KT. This would be inaccurate, and you should not use this model to forecast wind values based on the large difference between the observed and model data.

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Wind Dir	NE	NE	E	E	E	E	E	E	E	E	E	E	E	E	E	NE	NE	NE	NE	NE	NE	NE	NE	NE
Gust																								

SPECI KBIX 031602Z 03006KT 10SM SCT014 06/01 A3020 RMK A02A SLP230 T00560009

Time: Ensure same time when comparing data.

Model – 1100L / 1600Z

Observation – 1602Z / 1102L

Temperature: Convert F to C. Round values to nearest degree.

Accurate = 0C Difference

Nearly Accurate = 1-2C Difference

Inaccurate = 3C+ Difference

Model – 54F / 12C

Observation – 6C / 43F

Inaccurate: 6 Degree Difference

Dewpoint: Convert F to C. Round to nearest degree.

Accurate = 0C Difference

Nearly Accurate = 1-2C Difference

Inaccurate = 3C+ Difference

Model – 35F / 2C

Observation – 1C / 34F

Nearly Accurate: 1 Degree Difference

Wind: Convert cardinal direction to degrees.

Accurate = Direction / Within cardinal direction range. Speed / 3KT or less Difference

Nearly Accurate = Direction / NA. Speed / 4-6KT Difference

Inaccurate = Direction / Outside of cardinal direction range. Speed / 7KT+ Difference

Model – NE @ 5KT

Observation – 03006KT / NE @ 6KT

Accurate: Direction Within Range. Speed 1KT Difference.

Practice: Model Verification

PC: Model Verification